

Magical Orbs

There are n magical orbs, the i -th orb has magical power a_i . Evirir the dragon wizard wishes to combine all orbs to one super mega ultra magical orb. To do so, Evirir the dragon can do the following operation while there is more than one magical orb:

- Pick two orbs with power x and y .
- Destroy the two orbs and make a new orb with magical power $x + 2y$.

As Evirir the dragon did not graduate from wizard school, he needs your help to find the largest magical power the combined orb can have.

Input Format

There are multiple test cases. The first line contains the number of test cases T . Each of the next T lines contains a test case. The description of a test case follows.

The first line of each test case contains a single integer n . The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^{18}$) denoting the powers of the orbs.

It is guaranteed that the sum of n over all testcases is at most $2 \cdot 10^5$.

Output Format

Output T lines. For each test case, output the biggest magical power you can achieve modulo $10^9 + 7$.

Note: Modulo means “taking the remainder of”, you have to output the remainder of the answer after dividing by $10^9 + 7$.

Constraints

Task 1: samples (0 points) Task 2: $n, T \leq 10$ (10 points) Task 3: $a_1 = a_2 = \dots = a_n$ and $a_i \leq 10^9$ (10 points) Task 4: $a_i \leq 10^9$ (20 points) Task 5: no constraints (20 points) Task 6: no constraints (20 points) Task 7: no constraints (20 points)

Sample Input

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2
3
1 2 1
4
1 2 3 2
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Sample Output

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11
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Sample Explanation

In the first test case, we merge orb 3 (power $x = 1$) and orb 2 (with power $y = 2$), making a new orb with power $x + 2y = 1 + 2 \cdot 2 = 5$. Then, we can merge orb 1 (with power $x = 1$) with the new orb (with power $y = 5$) to get a new orb with power $1 + 2 \cdot 5 = 11$.